

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
df=pd.read_csv(r"C:\Users\dhruv\Downloads\1. hr_analytics.xls")
df
```

	Age	Gender	Department	MonthlyIncome	TotalWorkingYears	YearsAtCompany	JobSatisfaction	WorkLifeBalance	OverTime	
0	50	Female	HR	4753	14	8	2	1	Yes	
1	36	Male	R&D	13123	3	7	3	4	No	
2	29	Male	HR	19376	16	15	1	3	Yes	
3	42	Male	HR	9588	19	14	2	4	No	
4	40	Male	R&D	9304	18	8	2	3	No	
...	...	...	...	...	...	...	...	...	...	...
995	34	Female	Sales	7986	2	5	4	3	Yes	
996	51	Female	Sales	4481	26	12	4	2	Yes	
997	44	Male	HR	6731	27	6	4	1	No	
998	40	Female	Sales	19835	27	9	1	2	No	
999	53	Female	HR	13758	15	1	2	4	No	

1000 rows × 10 columns

```
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

```
Rows: 1000
Columns: 10
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   1000 non-null  int64
1   Gender                1000 non-null  object
2   Department            1000 non-null  object
3   MonthlyIncome         1000 non-null  int64
4   TotalWorkingYears     1000 non-null  int64
5   YearsAtCompany        1000 non-null  int64
6   JobSatisfaction       1000 non-null  int64
7   WorkLifeBalance       1000 non-null  int64
8   OverTime              1000 non-null  object
9   Attrition             1000 non-null  object
dtypes: int64(6), object(4)
memory usage: 78.3+ KB
```

```
df.isnull().sum()
```

```
Age                0
Gender             0
Department         0
MonthlyIncome      0
TotalWorkingYears  0
YearsAtCompany     0
JobSatisfaction    0
WorkLifeBalance    0
OverTime           0
Attrition          0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
total_employees = len(df)
print(f"There are {total_employees} employees in the dataset.")
```

There are 1000 employees in the dataset.

```
num_departments = df['Department'].nunique()
print(f"There are {num_departments} departments in the dataset.")
```

There are 3 departments in the dataset.

```
all_departments = df['Department'].unique()
print(all_departments)
```

```
['HR' 'R&D' 'Sales']
```

```
average_age = df['Age'].mean()
print(f"The average employee age is: {average_age:.1f} years old")
```

The average employee age is: 40.9 years old

```
min_age = df['Age'].min()
max_age = df['Age'].max()
print(f"Youngest employee age: {min_age} years old")
print(f"Oldest employee age: {max_age} years old")
```

Youngest employee age: 22 years old
Oldest employee age: 59 years old

```
gender_counts = df['Gender'].value_counts()
print(gender_counts)
```

```
Gender
Male      504
Female    496
Name: count, dtype: int64
```

```
department_counts = df['Department'].value_counts()
print(department_counts)
```

```
Department
HR          352
Sales       343
R&D         305
Name: count, dtype: int64
```

```
(df["TotalWorkingYears"]>10).sum()
```

```
np.int64(720)
```

```
df.groupby("Department")["MonthlyIncome"].mean()
```

```
Department
HR          11604.042614
R&D         11461.272131
Sales       11417.247813
Name: MonthlyIncome, dtype: float64
```

```
df[df["Attrition"] == "Yes"].groupby("Department").size()
```

```
Department
HR          70
R&D         74
Sales       59
dtype: int64
```

Start coding or [generate](#) with AI.

